3-Week Reading Plan

Frantz Final Reading Assignment: Hadrons/Hadronization/Jets

PHYS 7502 – Paced Study Schedule

How to Use This Plan

The final reading assignment from Prof. Frantz is dense – it pulls together material from at least five separate sources (his own slides, Vaidya's UCL jet slides, an arXiv review paper, a Webber conference talk, and multiple chapters of Wong's textbook) and assumes you can knit them together. This plan breaks the assignment into a realistic 3-week schedule with specific daily targets.

The Two-Pass Strategy

The most efficient way to get through dense material you find fast:

1. **First pass:** Read my corresponding study PDF (Part 1, Part 2, etc.) – these are designed as slow, pedagogical build-ups.
2. **Second pass:** Read Frantz's actual assigned sources. With the framework in place, the original reads much faster.

Time estimates below assume you're doing both passes. If you skip the first pass, the original reading takes 50% longer and you retain less. Don't skip.

Total Time Commitment

- Week 1: ~ 4 hours (foundation + experimental jets)
- Week 2: ~ 5 hours (theoretical framework)
- Week 3: ~ 5 hours (advanced models + wrap-up)

Total: ~ 14 hours spread across 3 weeks ($\approx$ 4–5 hours/week, 5 days/week). Pace yourself – this is a marathon, not a sprint.

All page numbers below refer to Frantz's assignment PDF (`phys7502_frantz_Spring_final_assignment_read`). "Part N" refers to my study PDF; these are the scaffolding you should read *first* each day.

1 Week 1: Foundation and Experimental Side

Goal for the week: Understand why hadronization is hard, what jets are, how we find them, and how event generators bridge theory and experiment. By the end of the week, you should be able to explain to a friend what an anti-k_T jet is and why we use it.

1.1 Day 1 – Framing the Problem

Day 1 Reading

Time: 45 minutes

First: Read my **Part 1** PDF (*The Big Picture*, 6 pages). This gives you the conceptual foundation for everything that follows.
Then: Read pp. 1–3 of Frantz's assignment PDF:

- p. 1: title slide
- p. 2: "Hadrons/Hadronization/Jets" overview slide
- p. 3: "Hadron Models" slide

Day 1 Self-Check

By end of day, you should be able to answer:

- Why can't we calculate hadronization from perturbative QCD?
- What does lattice QCD give us, and where does it fall short?
- What's the difference between structural models, classification models, and statistical models?

1.2 Days 2–3 – Jets at the LHC

Days 2–3 Reading

Time: 2.5 hours (split across 2 days; roughly 1.25 hr/day)

First: Read my **Part 2** PDF (*Jets at the LHC*, 10 pages) – the longest of my PDFs and the one with the most experimental content. Go slowly through the distance-formula section.
Then: Read pp. 4, 6–23 of Frantz's assignment:

- p. 4: "Start with Jets – Event Generators" slide
- pp. 6–13: Vaidya jet slides (why jets, what is a jet, IR safety, collinear safety, cone algorithms)
- pp. 14–18: Vaidya on modern jet algorithms (k_T, C/A, anti-k_T, the unified distance formula)
- pp. 20–23: Boosted jets, substructure, BDRS, trimming – **Frantz says "you may skim this section"** but the big picture is worth getting.

Optional deep dive: The full Vaidya slides are at
`https://www.hep.ucl.ac.uk/postgrad/teaching/lhc/Jets_2017.pdf`
(linked from Frantz p. 4). Skip this unless you want more detail.

Days 2–3 Self-Check

- Why are **IR safety** and **collinear safety** non-negotiable properties for a jet algorithm?
- Write down the generalized distance formula. What does each of $p = +1, 0, -1$ correspond to?
- Why do ATLAS and CMS almost exclusively use anti-k_T?
- For a boosted heavy object of mass m at transverse momentum p_{T}, what's the approximate angular separation of its decay products?

1.3 Day 4 – Event Generators

Day 4 Reading

Time: 45 minutes

First: Read my **Part 3** PDF (*Event Generators*, 6 pages). Short and mostly taxonomic – you're learning names and roles.
Then: Read p. 5 of Frantz's assignment:

- p. 5: "High Energy Event Generators" slide (PYTHIA, HERWIG, SHERPA, HIJING)

Truly optional references (only if you end up actually using these tools):

- PYTHIA homepage: `https://pythia.org/`
- arXiv:1901.04220 (HIJING details)
- arXiv:nucl-th/9502021 (original HIJING paper)

Day 4 Self-Check

- Name the four steps of event generation. Which is the non-perturbative one?
- Which event generator is used for heavy-ion collisions, and on top of what generator is it built?
- What's a "tune" in the PYTHIA context? Why are they experiment-specific?

1.4 Day 5 – Consolidation Day

Day 5 Activity

Time: 30–60 minutes

No new reading today. Instead:

- Re-read the summary pages of my Parts 1–3 (the "Deliverables" boxes at the end of each).
- Try to reproduce the unified distance formula from memory.
- Sketch the four-step event generator flowchart from memory.
- If any self-check question felt shaky, go back and re-read that section.

2 Week 2: Theoretical Framework

Goal for the week: Get the theoretical machinery that connects partons to hadrons in a factorizable way. Fragmentation functions, the Schwinger mechanism, the classical string model. This is the densest week – don't rush.

2.1 Days 1–3 – Fragmentation Functions

This is the hardest topic in the assignment. Budget accordingly.

Days 1–3 Reading

Time: 3 hours total ($\sim$ 1 hr/day)

Day 1 – Background:

- Read my **Part 4** PDF (*Fragmentation Functions*, 8 pages). Pay special attention to the x vs z convention warning.
- Read p. 24 of Frantz's assignment (his intro to FFs and the notation warning).
- Read p. 25 (Frantz's reading-source pointers).

Day 2 – Metz/Vossen review:

- Read pp. 26–28 of Frantz's assignment (excerpts from arXiv:1607.02521).
- Frantz says **"don't get bogged down, just scan."** Pay particular attention to:
  - The introductory paragraphs on what FFs are
  - The four types of FFs (integrated, TMD, higher-twist, di-hadron) – just know they exist
  - The factorization formulas for SIA, SIDIS, $pp \to hX$

Day 3 – Webber talk:

- Read pp. 29–41 of Frantz's assignment (Webber's conference proceedings article).
- Focus on:
  - Sections 1–2 (intro + theory of jet fragmentation)
  - Section 3 (hadronization models – especially the cluster vs string comparison)
  - Section 5 (**quark vs gluon jets** – Frantz flags this as the key takeaway)
- Skim Sections 4, 6–10 – they're interesting but not central.

Days 1–3 Self-Check

- Write down Webber's master factorization equation. What are C_i and D_i^h?
- In what sense do FFs have "scaling violations"? What equation governs their Q^2 evolution?
- What's the MLLA, and what characteristic experimental signature does it predict?

- What's the expected ratio of multiplicities in gluon jets vs quark jets in the asymptotic limit?

2.2 Day 4 – The Schwinger Mechanism

Day 4 Reading

Time: 1 hour

First: Read my **Part 5** PDF (*Schwinger Mechanism and Color Flux Tubes*, 8 pages). The *physical picture* is what matters; you can skim over the WKB derivation.
Then: Read pp. 42–47 of Frantz's assignment (Wong Ch. 5):

- p. 42: Frantz intro ("read first couple pages")
- pp. 43–44: §5.1 Schwinger particle production mechanism – **read carefully.** This sets up the color flux tube.
- p. 45: the potential setup (review only)
- pp. 45–47: detailed WKB tunneling calculation – **Frantz says "you may skim."** The physics endpoint is the result $P = \exp\left[-\pi(m^2 + p_T^2)/\kappa\right]$.

Day 4 Self-Check

- Why does the QCD color field between q and $\bar{q}$ form a flux tube rather than spreading out?
- What's the string tension κ? What value does it take? Where else will it appear?
- What two experimentally observed features fall out of the $e^{-\pi(m^2+p_T^2)/\kappa}$ formula?

2.3 Day 5 – Start on Classical String Model

Day 5 Reading

Time: 1 hour

First: Read **sections 1–3 of my Part 6** PDF – that is, the historical setup (dual resonance model), Regge trajectories, and the rotating-string derivation of the Regge slope. This covers roughly half of Part 6.
Then: Read pp. 48–51 of Frantz's assignment (Wong Ch. 7):

- p. 48: intro to string model
- p. 48–49 (top): §7.1 String Model of Hadrons
- pp. 50–51: rotating string, Regge slope derivation

You'll finish the rest of Wong Ch. 7 (yo-yo + Lund) next week.

Day 5 Self-Check

- Plot J vs M^2 for hadron resonances – what do you get?
- Derive the string prediction $J = M^2/(2\pi\kappa)$ starting from a rotating relativistic string. (This is a great homework-style problem.)
- From $\alpha' \approx 1\,\mathrm{GeV}^{-2}$, infer κ.

3 Week 3: Advanced Models and Wrap-Up

Goal for the week: Complete the Lund model, add the Reggeon/Pomeron framework for soft scattering, and close with the MIT bag model and its path to QGP. By week's end, you should be ready to answer Frantz's quiz questions and have a holistic view of the entire assignment.

3.1 Day 1 – Finish the Classical String and Lund Model

Day 1 Reading

Time: 1 hour

First: Read **sections 4–8 of my Part 6** PDF (yo-yo model, symmetric Lund model, the splitting function, gluon kinks).
Then: Read pp. 52–57 of Frantz's assignment:

- pp. 52–53: §7.2 Yo-yo model of stable particles – **read carefully.**
- p. 54: §7.3 Yo-yo state in motion – **Frantz says "you can probably skip this section."**
- pp. 54–55 (top): §7.4 Symmetric Lund model intro – "skim this one again."
- pp. 55–56: Lund splitting function derivation – "just skim these pages." Come away with equation (7.35)/(7.36), the Lund function.
- p. 56: Exercise 7.1 – Frantz marks this **"starts here is an exercise you can skip."**
- p. 57: **"Start here only till end of 7.4"** – this is the punchline on independent fragmentation.

Day 1 Self-Check

- Sketch the space-time picture of a yo-yo at rest. Find the period.
- Why does the Lund assumption of constant proper time lead to a flat rapidity distribution?
- Write down the Lund symmetric splitting function. What are the parameters a and b?

3.2 Days 2–4 – Dual Parton Model and the Pomeron

Days 2–4 Reading

Time: 2.5 hours total ($\sim$ 50 min/day)

Day 2 – Context and setup:

- Read **sections 1–2 of my Part 7** PDF (historical context, Veneziano amplitude).
- Read pp. 58–60 of Frantz's assignment:
  - p. 58: Frantz DPM intro

  - p. 59: Frantz's Pomeron/Reggeon review slide (**"read it again"** per his note)
  - p. 60: Color Dipole Framework digression

Day 3 – Wong on DPM core:

- Read **sections 3–4 of my Part 7** PDF (Pomeron, DPM multi-chain picture).
- Read pp. 61–65 of Frantz's assignment (Wong Ch. 8):
  - pp. 61–62: §8.1 Dual Resonance Model, Veneziano amplitude – read the main text
  - **"Skip Supplement 8.1 and 8.2"** (formal derivations)
  - p. 65: "starts here again" – return to the main text for the crossing-symmetric amplitude
  - **"Skip Supplement 8.3"**
  - p. 65 bottom: eq. 8.9, $\sigma_{\text{tot}} \propto s^{\alpha_0 - 1}$ – Frantz flags as **"key signature of the Reggeon/Pomeron behavior."**

Day 4 – Pomeron exchange and DPM-FF connection:

- Read **sections 5–6 of my Part 7** PDF.
- Read pp. 65–66 of Frantz's assignment:
  - p. 65 bottom: §8.2 Pomeron Exchange – **"this whole last page is quite important."**
  - p. 66: Frantz notes **"skim pp 150–151"** of Wong – this shows the DPM/FF connection.

Days 2–4 Self-Check

- What quantum numbers does the Pomeron carry? Why are those the right ones to explain forward diffractive scattering?
- Starting from Reggeon exchange $A \sim s^{\alpha(t)}$ and the optical theorem, derive $\sigma_{\text{tot}} \propto s^{\alpha_0 - 1}$.
- Why do high-energy pp total cross sections stay nearly constant with energy? What Pomeron intercept does this imply?

3.3 Day 5 – MIT Bag Model and QGP

Day 5 Reading

Time: 1.5 hours

First: Read my **Part 8** PDF (*MIT Bag Model and QGP*, 9 pages). Pay particular attention to the T_c derivation – Frantz loves this.
Then: Read pp. 67–78 of Frantz's assignment:

- p. 67: Frantz intro to bag model

- pp. 68–73: Slideplayer bag model slides (schematic intro, can read quickly)
- pp. 74–75: Wong §9.2 Bag Model of a Hadron – **read carefully.** This is where $p^0 R = 2.04$, $B^{1/4} \approx 206\,\text{MeV}$, and the equilibrium-radius argument come from.
- p. 76: §9.3 Quark-Gluon Plasma + §9.3.1 at high T
- p. 77: **Frantz says "ignore Supplement 9.2"** (detailed statistical mechanics)
- p. 78: **"just skim this one, . . . the punchline is the plot below"** – the QCD phase diagram.

Day 5 Self-Check

- Derive the bag-model equilibrium condition and find R for a given B and N.
- Given $B^{1/4} = 206\,\text{MeV}$, what's the predicted T_c for QGP? Compare to lattice QCD's $\sim 156\,\text{MeV}$.
- What's the physical picture of deconfinement in the bag model?

3.4 Day 6 – Review and Integration

Day 6 Activity

Time: 1 hour

No new reading. Instead:

- Read the "Series Summary" at the end of my Part 8.
- Try to draw, on a single sheet, the conceptual flowchart of the whole assignment: perturbative QCD → jets → factorization/FFs → string picture → DPM/Pomeron → bag/QGP.
- Go back and answer any self-check questions you were shaky on.
- If Frantz's quiz questions for this assignment have been distributed, attempt them.

4 Master Reference: Sources in Frantz's Assignment

For quick navigation. Pages refer to Frantz's assignment PDF.

Pages	Source	Focus
1–5	Frantz intro slides	Overview, hadron models, generators
6–23	Vaidya (UCL) slides	Jet algorithms, substructure
24–25	Frantz FF intro	Convention warning, reading list
26–28	Metz & Vossen (arXiv:1607.02521)	Types of FFs, factorization
29–41	Webber talk	Theory of jet fragmentation, models
42–47	Wong Ch. 5 (Schwinger)	Pair production, flux tubes
48–57	Wong Ch. 7 (String)	Regge, yo-yo, Lund model
58–66	Wong Ch. 8 + Frantz	Dual Parton Model, Pomeron
67–73	Slideplayer + Frantz	Bag model intro
74–78	Wong Ch. 9	Bag model + QGP

5 Cross-Reference: My PDFs ↔ Frantz's Pages

My PDF	Topic	Frantz pp.	Week
Part 1	Framing / big picture	1–3	W1 D1
Part 2	Jets at the LHC	4, 6–23	W1 D2–3
Part 3	Event generators	5	W1 D4
Part 4	Fragmentation functions	24–41	W2 D1–3
Part 5	Schwinger mechanism	42–47	W2 D4
Part 6	String / Lund model	48–57	W2 D5 + W3 D1
Part 7	DPM, Regge, Pomeron	58–66	W3 D2–4
Part 8	Bag model + QGP	67–78	W3 D5

6 Study Habits Worth Adopting

Practical Tips

1. **Read actively with a pen.** On hard derivations (Regge slope, T_c, Lund splitting function), actually rederive on scratch paper as you read. Passive reading doesn't stick.
2. **Keep a "one-pager" running**. On a single sheet of paper, write down: key equations (one per topic), key numbers (κ, $B^{1/4}$, T_c, α'), and the unifying thread ($\kappa \approx 1\,\mathrm{GeV/fm}$ everywhere).
3. **Follow Frantz's emphasis.** He marked what to read carefully, skim, and skip. Trust him. Anything marked "important," "key signature," or "read again" is likely to show up on the quiz.

4. **Don't fight notation**. The x vs z fragmentation convention, the p parameter in the jet distance formula, the α intercepts – different sources use different conventions. Note these as you go rather than letting them confuse you.

5. **Skip the math you're told to skip**. Frantz marks several supplements (8.1, 8.2, 8.3, 9.2) as skippable. The physics lives in the main text; the supplements are formal derivations you can return to only if needed.

7 If You Fall Behind

Triage Plan

If life happens and you're short on time, here's the priority order:

1. **Must do**: Parts 1, 2, 4, 6, 8. These cover the physics of jets, fragmentation, the string model, and bag/QGP. Together they get you through 80% of likely quiz material.

2. **Should do**: Parts 5 and 7. Schwinger is short; DPM/Pomeron is conceptually nice but less likely to be heavily tested.

3. **Can skip in emergencies**: Part 3 (event generators – mostly nomenclature you can pick up in context).

Better to understand six parts well than to rush through all eight superficially.

Final Note

This reading is worth doing carefully. The material in this assignment is *foundational* for anyone working on experimental or phenomenological QCD – jets, hadronization, PYTHIA, Pomerons, QGP, these all come up constantly in modern high-energy physics. You'll return to this framework repeatedly in future coursework and research, so the time you invest now pays dividends later.

Good luck with the reading.